Ayush Singh

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Education

- **Pandit Deendayal Energy University** Gandhinagar, India
  Bachelor of Technology in Mechanical Engineering Engineering, CPI: 9.88/10 (WES GPA: 3.97/4) *2023 - Present*

Skills

CFD (Python/MatLab, Ansys Fluent, OpenFOAM basics), FEA (Ansys Mechanical), CAD (SolidWorks, NX, Fusion 360), Welding (GTAW, GMAW, LASER, Ultrasonic, Friction, FSW, Resistance, Plasma, Gas), CNC Turning (Siemens), Advanced Manufacturing (EDM, WAAM, LPBF), MatLab, Model Rocketry, Programming & Full Stack Web Development, LaTeX

Experience

- **Undergraduate Research Fellow & Summer Intern - IIT Gandhinagar** May 2025 - Present
  Prof. Manish Kumar, IIT Gandhinagar
  - **Role Progression**: Transitioned from Summer Intern to Research Fellow to design and execute physical experiments for computational model validation.
  - **Award**: 1st prize for Summer Research Excellence (SRE) for outstanding research on snow avalanche modeling.
  - **CFD Simulation**: Granular flow using VOF in Ansys Fluent; validated with experimental data (<5% error) (Report Link).
  - **Structural Testing**: Simulated 2 lattice structures; reduced impact force by up to 30%.
- **Founder, Advisor & Systems Engineer - Team Zenith & Treasurer at Apogee3** Sep 2023 - Present
  Model Rocketry Team & Aerospace society of PDEU
  - **Foundation & Funding**: Founded the university model rocketry team and successfully secured ₹3,50,000 ($3,600) in corporate and institutional sponsorships to finance vehicle manufacturing and competition logistics.
  - **Global Ranking (Advisor)**: Advising the team through the Teknofest Rocket Competition (Turkey), achieving 8th rank globally (3rd in India) following the Preliminary Design Review (PDR).
  - **National Finalist**: CanSat-deploying model rocket (1 km apogee) with dual-recovery systems, securing finalist status and an approved Flight Readiness Review (FRR) at the IN-SPACe National Competition as Systems, Design & Analysis engineer.
  - **Rocket Motor**: Tested different rocket motors with varied dimensions and propellants improving thrust-to-mass ratio by over 90% by employing metal casing motor with converging-diverging nozzle
  - **Events**: Conducted 15+ events in the 3 years including workshops and cultural events

Projects / Research Work

- **Novel design approach to reduce pressure drop & increase heat transfer rate in TPMS-based heat exchangers**
  Prof. Pavan Gurrala, PDEU
- **Optimized design of lattice-based structures to reduce avalanche impact by means of CFD modeling**
  Prof. Manish Kumar, IIT Gandhinagar
- **Shape Memory Alloy actuated hand exoskeleton for stroke rehabilitation**
  Dr. Nilavjyoti Sarmah, Dr. Salman Khan, PDEU
- **Optimization of fan speed in IIT-Gn hostel room (J131) when AC turned on**
  IIT Gandhinagar
- **Real Time Monitoring System for LPBF and GTAW Systems**
  Dr. Ojas Satbhai, PDEU
  Novel approach to capture acoustic data for crack mitigation in bio-compatible ZK60 alloy for orthopedic implants

Publications

- **3D printed Mandibular Reconstruction without bone graft: implant-only procedure**
  Prof. Pavan Gurrala, PDEU, Dr. Girish Amlani, Zydus hospital - Under Review - ICRAMIE 2027, NIT Warangal
- **Parameter Sensitivity of the Three-Element Windkessel Model for Vascular Stenting**
  Dr. Ojas Satbhai, PDEU, Dr. Jnana Ranjan Senapati, NIT Rourkela - Accepted - FMFP 2026, NIT Rourkela
- **Physics-Aligned Modelling and Multi-Objective Fitness Function Development for a Batch RO System**
  Dr. Rahul Deharkar, PDEU - Accepted for publication - IMECE 2026, ASME
- **Property and Composition Dependence of hot cracking in LPBF for different materials**
  Prof. Vishvesh Badheka, PDEU - Published (doi.org/10.1115/IMECE-INDIA2025-161260) - IMECE 2025, ASME
- **Predictive Analysis of Multi-Effect-Distillation System using ANN**
  Dr. Rahul Deharkar, PDEU - Accepted for publication - ICAWTM-25 (Springer)
- **Additive Manufacturing of Batteries: Recent Trends and Challenges**
  Dr. Ojas Satbhai, PDEU - Under Review - ICTEA
- **High-Precision Launch Mount for Small Rockets Featuring Gear-Driven Angular Control**: Design Patent Under Review
- **Aircraft Wing Spar comprising Honeycomb I-beam with supports**: Design Patent Under Review

Achievements

- Bhalodia-Khetan Summer Research Excellence Award, ₹50,000 ($570) prize, 1st prize at SRIP, IIT Gandhinagar, 2025
- Merit Scholarship Recipient (100% Tuition Fee Waiver) based on GPA and JEE, 2023-2027
- University Topper across all semesters (Director's Appreciation prize every year), PDEU, 2023-2026
- Selected at Prestigeous Student Research Internship Program (0.3% acceptance rate) at IIT Gandhinagar, 2025
- 3rd Rank, Mech-A-Thon, Optimized wing spar design for aircraft and achieved 16% weight reduction, 2025
- 10-day Residential Model Rocketry Training (amongst 70 students from across India), Bengaluru (IN-SPACe, ISRO), 2024
- Consolation Prize at IEEE SPS Hackathon GS MSP CUP, Prediction of schizophrenia, 2024
- Invited Speaker at Additive Manufacturing Young Professional Seminar, 2024
- Panelist at "Science Festival – Make, Innovate, Lead", Respire, GUJCOST, 2021
- 3rd Winner in National COVID-19 Idea Challenge on 'AI & Robotics', VASCSC, 2020
- Finalist, Children Innovation Festival - 2k19, GUSEC, 2019

Certifications

Flight Mechanics (IIT Bombay) Topper, Metal 3D printing (Michigan), Launch Vehicle Analysis & Design (IIT Bombay), Essentials of Model Rocketry (IN-SPACe, ISRO), Java SE 8 Fundamentals (Oracle)

Leadership / Volunteering Involvement

- Astronomy Outreach via Instagram: @atroyush: Created and actively manage a dedicated Instagram account for astronomy.
- Space Science content Developer for website and outreach, Intern at Tanmaye's Amazing Space
- Technical Trainer for C, C++ and Game Development with Unity, Intern at Royal Technosoft Pvt. Ltd.
- Lawn Tennis Secretary, PDEU Sports Committee (6+ events, national participation of 100+ people), 2024 - 2026
- Climate change and waste management awareness campaign in government school, Rotary Club of Shantigram Ahmedabad & Khodiyar village local government (NGO Volunteering)
- SciKnowTech Foundation (volunteering for STEAM outreach and events with more than 500 people involved)
- Coordinated 2 CFD (FDM, FVM) Python-based workshops at PDEU, to train 100+ BTech, MTech, and PhD students
- Lead Coordinator, 1st International Battery Symposium (ICTEA 2024) with McMaster, Toronto Met, Yalova Univ. & PDEU
- Founding President - SciKnowTech VIPNET Club, science club registered under government of India, 2021 - 2023
- Active member of professional bodies: American Society of Mechanical Engineers (ASME), Indian Institute of Welding (IIW), Society of Automotive Engineers (SAE)
- School Vice President (After elevated from Deputy Minister and Minister positions) - handling leadership roles and system affairs ministry in the school parliament, 2018 - 2023

Interests

- **Technical Interests**: Computational Fluid Dynamics, Welding, Non-conventional & Additive Manufacturing, Astronomy, Model Rocketry and Aerospace, Design, Thermal, Structures, Granular Mechanics
- **Personal Interests**: Fine Arts and Painting, Sports (Lawn Tennis, Karate), High Altitude Trekking

Test Scores

- **Graduate Record Examination (GRE)**: 317 (170 Q + 147 V) + 3 AWA
- **International English Language Testing System (IELTS)**: 8.5 Band (L9 R9 S8 W7.5)
- **Aerospace Engineering Graduate Aptitude Test in Engineering (GATE 2025)**: 26/100 (745 Rank - 88.70 %ile)
- **Mechanical Engineering Graduate Aptitude Test in Engineering (GATE 2025)**: 25.67/100 (8023 Rank - 86.54 %ile)
- **Joint Entrance Examination (JEE) Mains**: 109/300 (71364 Rank - 93.68 %ile)

References

Prof. Manish Kumar (Research Guide, Dean of Student Affairs, IIT Gn) - mkumar@iitgn.ac.in
Prof. Pavan Gurrala (Research Guide, Associate Dean of School, PDEU) - Pavan.Gurrala@sot.pdpu.ac.in
Prof. Vishvesh Badheka (Research Guide, Former Dean of Academic Affairs, PDEU) - Vishvesh.Badheka@spt.pdpu.ac.in
Dr. Anirudh Kulkarni (Faculty Mentor & CFD Guide) - Anirudh.Kulkarni@sot.pdpu.ac.in